What Is in a Name? Towards New Definitions of Superintelligence and Their Implications for AI Risk Discourse

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Motivation: Naming Is Not Neutral

- The word “superintelligence” carries built-in assumptions
- Intelligence as a single, measurable quantity
- A sufficiently capable system as singular and discrete
- Arrival as a discontinuous break with the present
- These assumptions shape policy, investment, and public fear

The Problem With the Standard Framing

- Dominant accounts: Bostrom's recursive self-improvement; Kurzweil's exponential singularity
- Popular imagination: superintelligence as secular deity, or secular apocalypse
- We argue this framing is not just contestable, but practically dangerous

AI Is Already Embedded in Civilisation

- Credit decisions in financial markets
- Yield optimisation in agricultural supply chains
- Clinical decision support in healthcare
- Information flows in democratic governance

The real risk: systemic fragility, not a singular future entity

Three Contributions of This Paper

1. Review entity-centric definitions of superintelligence and their assumptions
2. Propose two alternative definitions: systemic and behavioural
3. Show how these redefinitions reshape AI risk, governance, and alignment

The Entity-Centric Definition

- Bostrom: an intellect greatly exceeding human cognition in virtually all domains
- Kurzweil: the Singularity as the moment AI surpasses human intelligence
- Shared structure: a threshold, a discrete system crossing it, transformative consequences

Limits of the Entity-Centric View

- Assumes intelligence is unidimensional and aggregable
- Assumes a binary threshold, but capability growth has been uneven
- Directs attention to a hypothetical future event
- Diverts attention from present, growing systemic risks

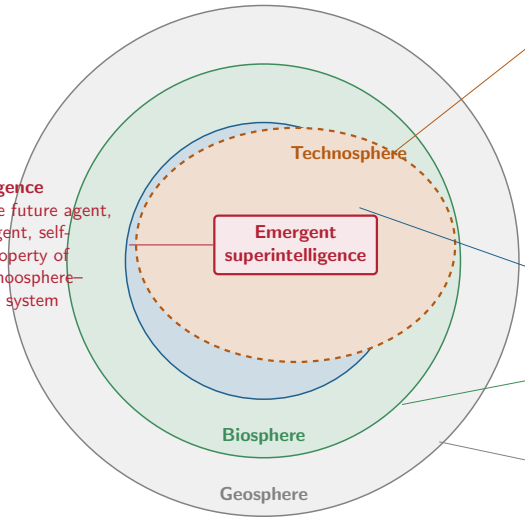
The Hidden Cost of Hype

- The singularity framing generates attention, investment, and narrative drama
- It may distort investment allocation and fuel hype cycles
- Unmet expectations can trigger confidence collapse and AI winters
- With AI now embedded in finance, agriculture, and healthcare, an AI winter carries systemic, not just reputational, risk

Alternative 1: A Systemic Definition

- Superintelligence as an emergent property of the *technosphere*
- The interconnected network of human agents, AI systems, institutions, and infrastructures
- Exhibits adaptive, self-organising behaviour no single component exhibits
- Something resembling superintelligence may already exist, as a present reality

From Biosphere to Technosphere



Technosphere
the globally coupled system of technology, infrastructure and institutions (Haff, 2014); now extends beyond and reshapes the noosphere that produced it

Superintelligence
not a discrete future agent, but an emergent, self-organising property of the coupled noosphere–technosphere system

Noosphere
the sphere of human thought, culture and cognition, arising from and transforming the biosphere (Vernadsky, Le Roy, Teilhard de Chardin)

Biosphere
the totality of living matter (Vernadsky, 1926)

Geosphere
rock, atmosphere, oceans — the abiotic substrate

Historical Resonance

- Fire, writing, and paper transformed civilisation
- Not through a single act of genius
- Through slow, distributed accumulation of practices and infrastructure
- Their effects were emergent, not singular

Implications of the Systemic Definition

- Risk shifts from a single rogue agent to systemic fragility and cascading failure
- Complexity science, financial regulation, and infrastructure governance already offer tools
- Alignment means aligning the network, not just individual AI systems
- Requires attention to incentives, distributional effects, and ecological sustainability

Alternative 2: A Behavioural Definition

- Focus on spontaneous goal-directed behaviour in multi-agent systems
- Goals that arise from agent interaction, not explicit programming
- May diverge from the intentions of any individual designer
- Treats goal-directedness as graded and observable, not a binary event

Hybrid Human-AI Societies

- Goals, norms, and strategies co-evolve through human-AI interaction
- Raises practical questions about the moral status of AI agents
- Bears on the social contracts governing human-AI interaction
- Existing superintelligence vocabulary is not well equipped for this shift

Implications for Risk and Governance

- Urgent risks lie in fragile, opaque socio-technical networks, not a future monolithic entity
- Roughly eight billion people live in economies entangled with AI-mediated systems
- Alignment requires coordinating incentive structures, institutions, and regulation
- Not just specifying a single system's objective function

Conclusions

- The name “superintelligence” matters, and current definitions distort risk analysis
- Systemic definition: an emergent property of complex socio-technical networks
- Behavioural definition: spontaneous goal-directed behaviour in multi-agent systems
- The urgent work is understanding, governing, and aligning the system we already have

Summary: The Argument at a Glance

